A New Decomposition of the Divisor Summatory Function and Its Asymptotic Formula

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Abstract. In this paper, we study the asymptotic behavior of the divisor summatory function $D(x)$. We establish the exact identity

$$D(x) = 1 + \sum_{i\geq 1}(i+1)\pi(x^{1/i}) + \sum_{j\geq 2}(-1)^j(j-1)T_j(x),$$

which decomposes $D(x)$ via inclusion–exclusion according to the number of distinct prime factors of n. The three terms correspond respectively to $n = 1$, prime powers, and integers with $\omega(n) \geq 2$, where the last sum constitutes the main term.

As the first step toward analyzing $D(x)$, we establish an asymptotic formula with an explicit error term for $T_j(x)$ with fixed $j \geq 2$. We consider the two-variable Dirichlet series $F(s, y)$ and obtain the factorization $F(s, y) = \zeta(s)^{2+2y}H(s, y)$, where H is holomorphic near $s = 1$. Applying the Selberg–Delange method—a truncated Perron formula, contour deformation, and local Hankel analysis—we prove that

$$T_j(x) \sim \frac{2^j}{j!}\, x\log x\,(\log\log x)^j \quad \text{as } x \to \infty.$$

We hope to extend our results to obtain an asymptotic formula for $D(x)$ itself.

1 Introduction

1.1 The Dirichlet divisor problem

Let $d(n)$ denote the number of divisors of a positive integer n, and let

$$D(x) := \sum_{n\leq x} d(n) \tag{1.1}$$

be its summatory function. Dirichlet [] proved in 1849 that

$$D(x) = x\log x + (2\gamma - 1)x + \Delta(x), \tag{1.2}$$

where γ is the Euler–Mascheroni constant and $\Delta(x) = O(x^{1/2})$. The Dirichlet divisor problem asks for the infimum θ of all real numbers such that $\Delta(x) = O(x^{\theta+\varepsilon})$ for every $\varepsilon > 0$.

Hardy [] proved in 1916 that $\theta \geq 1/4$, and it is conjectured that $\theta = 1/4$. In the other direction, Voronoï [] first improved Dirichlet's bound to $\theta \leq 1/3$ in 1904, and subsequent refinements based on Vinogradov's method of exponential sums [] led to the current best estimate, due to Huxley [] in 2003:

$$\theta \leq 131/416. \tag{1.3}$$

In this paper, rather than pursuing further improvements of θ, we explore an alternative approach: an exact combinatorial decomposition of $D(x)$ via inclusion–exclusion on the number of distinct prime factors of n.

1.2 A new decomposition

For a positive integer n, let $\omega(n)$ denote the number of distinct prime factors of n. For each integer $j \geq 2$, define

$$T_j(x) := \sum_{n \leq x} d(n) \binom{\omega(n)}{j} \tag{1.4}$$

where $\binom{\omega(n)}{j}$ is the binomial coefficient. The main identity of this paper is the following.

Theorem 1.1. For every real number $x \geq 1$,

$$D(x) = 1 + \sum_{i \geq 1} (i+1)\pi(x^{1/i}) + \sum_{j \geq 2} (-1)^j (j-1) T_j(x) \tag{1.5}$$

where $\pi(x)$ denotes the prime-counting function.

The three terms on the right-hand side of (1.5) correspond, respectively, to $n = 1$, prime powers $\omega(n) = 1$, and integers with $\omega(n) \geq 2$. The proof of Theorem 1.1 is given in Section 2.

Classical results of Dirichlet give the asymptotic $D(x) \sim x \log x$. By the prime number theorem, the prime-power contribution satisfies

$$\sum_{i \geq 1} (i+1)\pi(x^{1/i}) \sim \frac{2x}{\log x}, \tag{1.6}$$

which is of strictly smaller order than $x \log x$. Hence the first two terms in the identity are negligible compared with the main asymptotic of $D(x)$, and the third sum

$$\sum_{j \geq 2} (-1)^j (j-1) T_j(x) \tag{1.7}$$

must constitute the main term of $D(x)$. Individually, each $T_j(x)$ grows rapidly with j, but the alternating coefficients $(-1)^j(j-1)$ induce substantial cancellation, reducing the total contribution to the order $x \log x$. Understanding the asymptotic behavior of each $T_j(x)$ is therefore the first step toward an asymptotic formula for $D(x)$ itself.

Example 1.2. At $x = 10000$, direct computation gives

$$T_2(10000) = 297{,}078 \qquad T_3(10000) = 140{,}640$$
$$T_4(10000) = 26{,}674 \qquad T_5(10000) = 1{,}216$$

and $T_j(10000) = 0$ for $j \geq 6$. The prime-power contribution equals 2,711, and the identity (1.5) yields

$$1 + 2{,}711 + (T_2 - 2T_3 + 3T_4 - 4T_5) = 1 + 2{,}711 + 90{,}956 = 93{,}668 = D(10000),$$

confirming Theorem 1.1 numerically. Moreover, the ratio between $T_j(x)$ and its conjectured asymptotic $\frac{2^j}{j!} x \log x (\log\log x)^j$ converges to 1 only very slowly: at $x = 10^9$, this ratio is 0.43 for $j = 2$ and 0.21 for $j = 3$. This slow convergence motivates the refined asymptotic expansion of $T_j(x)$ developed in Section 6.

1.3 Main results

To analyze $T_j(x)$, we consider the two-variable Dirichlet series

$$F(s,y) := \sum_{n\geq 1} \frac{d(n)(1+y)^{\omega(n)}}{n^s}, \tag{1.8}$$

which encodes $T_j(x)$ as the coefficient of y^j in the generating relation $\sum_{j\geq 0} T_j(x) y^j$. Here, we extend the definition of $T_j(x)$ to all $j \geq 0$ by the formula (1.4), so that $T_0(x) = D(x)$ and $T_1(x) = \sum_{n\leq x} d(n)\omega(n)$. The series $F(s,y)$ admits the factorization

$$F(s,y) = \zeta(s)^{2+2y} H(s,y), \tag{1.9}$$

where $H(s,y)$ is holomorphic in s on a neighborhood of $\mathrm{Re}(s) \geq 1$ and uniformly bounded for y in a small disk around 0. This factorization isolates the singularity of $F(s,y)$ at $s=1$, and it is the starting point of the Selberg–Delange method.

Our main asymptotic result is the following.

Theorem 1.3. For each fixed integer $j \geq 2$, as $x \to \infty$,

$$\begin{aligned} T_j(x) = {} & \frac{2^j}{j!}\, x \log x\, (\log\log x)^j \\ & + \frac{2^{j-1}}{(j-1)!}\big(\partial_y H(1,0) - 2(1-\gamma)\big) x \log x\, (\log\log x)^{j-1} \\ & + O\big(x \log x\, (\log\log x)^{j-2}\big). \end{aligned} \tag{1.10}$$

In particular,

$$T_j(x) \sim \frac{2^j}{j!}\, x \log x\, (\log\log x)^j. \tag{1.11}$$

The proof of Theorem 1.3 is given in Sections 4–6. The essential ingredients are the truncated Perron formula, contour deformation around $s=1$, and a local Hankel analysis in the w-plane obtained via the substitution $w = (s-1)\log x$. The coefficient $\partial_y H(1,0)$ admits an explicit expression as an absolutely convergent series over primes; we record this expression in Section 6, where it is used in the proof of Theorem 1.3.

Combining Theorem 1.3 with the identity (1.5), we hope to obtain an asymptotic formula for $D(x)$ itself. We discuss this direction in Section 7.

1.4 Outline of the paper

The paper is organized as follows. In Section 2, we prove the identity (Theorem 1.1) by partitioning $D(x)$ according to $\omega(n)$ and applying an inclusion–exclusion argument to the composite part. In Section 3, we introduce the two-variable Dirichlet series $F(s,y)$ and establish its factorization $F(s,y) = \zeta(s)^{2+2y} H(s,y)$, together with the analytic properties of $H(s,y)$ needed in the sequel.

Sections 4–6 are devoted to the proof of Theorem 1.3. In Section 4, we apply the truncated Perron formula to the generating partial sum $A(x,y) := \sum_{n\leq x} d(n)(1+y)^{\omega(n)}$ and deform the contour around $s=1$. In Section 5, we carry out the local Hankel analysis via the substitution $w = (s-1)\log x$ and evaluate the main and lower-order contributions. In Section 6, we extract the asymptotic formula for $T_j(x)$ from that of $A(x,y)$ by differentiating in y, thereby completing the proof of Theorem 1.3.

Finally, in Section 7, we combine Theorem 1.3 with the identity (1.5) and discuss the remaining obstacles toward an asymptotic formula for $D(x)$.

2 Proof of the identity

In this section, we prove the identity stated in Theorem 1.1. The proof proceeds by partitioning the sum $D(x) = \sum_{n \le x} d(n)$ according to the value of $\omega(n)$, and then applying an inclusion–exclusion argument to the composite part.

Proof of Theorem 1.1. Partition the integers n with $1 \le n \le x$ into three disjoint classes:

(i) $n = 1$ (equivalently, $\omega(n) = 0$);

(ii) prime powers, i.e., $\omega(n) = 1$;

(iii) composite integers with $\omega(n) \ge 2$.

Writing $D(x)$ as the sum of contributions from each class, we have

$$D(x) = d(1) + \sum_{\substack{n \le x \\ \omega(n)=1}} d(n) + \sum_{\substack{n \le x \\ \omega(n) \ge 2}} d(n). \tag{2.1}$$

Since $d(1) = 1$, the first term equals 1.

For the second term, every integer n with $\omega(n) = 1$ is of the form $n = p^i$ for a unique prime p and a unique integer $i \ge 1$, and $d(p^i) = i + 1$. The number of prime powers $p^i \le x$ with exponent exactly i equals $\pi(x^{1/i})$. Hence

$$\sum_{\substack{n \le x \\ \omega(n)=1}} d(n) = \sum_{i \ge 1} (i+1)\pi(x^{1/i}). \tag{2.2}$$

For the third term, we apply inclusion–exclusion on the set of distinct prime divisors of n. For each integer $j \ge 2$, recall the definition

$$T_j(x) = \sum_{n \le x} d(n) \binom{\omega(n)}{j}.$$

We claim that

$$\sum_{\substack{n \le x \\ \omega(n) \ge 2}} d(n) = \sum_{j \ge 2} (-1)^j (j-1) T_j(x). \tag{2.3}$$

To prove (2.3), interchange the order of summation on the right-hand side:

$$\sum_{j \ge 2} (-1)^j (j-1) T_j(x) = \sum_{n \le x} d(n) \sum_{j \ge 2} (-1)^j (j-1) \binom{\omega(n)}{j}.$$

For n with $\omega(n) \in \{0, 1\}$, we have $\binom{\omega(n)}{j} = 0$ for all $j \ge 2$, so such n do not contribute. For n with $\omega(n) = m \ge 2$, we use the following combinatorial identity.

Lemma 2.1. For every integer $m \ge 2$,

$$\sum_{j=2}^{m} (-1)^j (j-1) \binom{m}{j} = 1. \tag{2.4}$$

Proof of the lemma. Starting from the standard identities

$$\sum_{j=0}^{m}(-1)^j\binom{m}{j}=0 \quad \text{and} \quad \sum_{j=0}^{m}(-1)^j\, j\binom{m}{j}=\frac{d}{dx}(1-x)^m\bigg|_{x=1}=0 \text{ (for } m\geq 2\text{)},$$

we combine them to obtain

$$\sum_{j=0}^{m}(-1)^j(j-1)\binom{m}{j}=0 \quad (\text{for } m\geq 2).$$

Separating the $j=0$ and $j=1$ terms:

$$1\cdot(-1)+0\cdot\binom{m}{1}+\sum_{j=2}^{m}(-1)^j(j-1)\binom{m}{j}=0,$$

which yields

$$\sum_{j=2}^{m}(-1)^j(j-1)\binom{m}{j}=1.$$

□

Applying the lemma, we obtain

$$\sum_{n\leq x}d(n)\sum_{j\geq 2}(-1)^j(j-1)\binom{\omega(n)}{j}=\sum_{\substack{n\leq x\\ \omega(n)\geq 2}}d(n)\cdot 1,$$

which establishes (2.3).

Combining the contributions from all three classes yields

$$D(x)=1+\sum_{i\geq 1}(i+1)\pi(x^{1/i})+\sum_{j\geq 2}(-1)^j(j-1)T_j(x),$$

completing the proof. □

3 Dirichlet series and factorization

In this section, we introduce the two-variable Dirichlet series $F(s,y)$ and establish its key analytic properties. The main output is the factorization $F(s,y)=\zeta(s)^{2+2y}H(s,y)$, where $H(s,y)$ is holomorphic in a neighborhood of $s=1$ and uniformly bounded in a small disk around $y=0$.

3.1 Euler product

Recall the definition

$$F(s,y):=\sum_{n\geq 1}\frac{d(n)(1+y)^{\omega(n)}}{n^s}. \tag{3.1}$$

Since the arithmetic function $n\mapsto d(n)(1+y)^{\omega(n)}$ is multiplicative in n for each fixed y, the series admits an Euler product representation for $\mathrm{Re}(s)>1$.

Lemma 3.1. For $\mathrm{Re}(s) > 1$ and every $y \in \mathbb{C}$,

$$F(s,y) = \prod_p \left(1 + (1+y)\left((1-p^{-s})^{-2} - 1\right)\right), \tag{3.2}$$

where the product is over all primes p.

Proof. By multiplicativity, for $\mathrm{Re}(s) > 1$ we have

$$F(s,y) = \sum_{n\geq 1} \frac{d(n)(1+y)^{\omega(n)}}{n^s} = \prod_p \left(\sum_{k\geq 0} \frac{d(p^k)(1+y)^{\omega(p^k)}}{p^{ks}} \right).$$

For $k = 0$, the summand equals 1. For $k \geq 1$, we have $d(p^k) = k+1$ and $\omega(p^k) = 1$, so the local factor becomes

$$1 + (1+y) \sum_{k\geq 1} \frac{k+1}{p^{ks}}.$$

Setting $u := p^{-s}$ and using the standard identity

$$\sum_{k\geq 1} (k+1)u^k = \frac{1}{(1-u)^2} - 1 \quad \text{for } |u| < 1,$$

the local factor equals $1 + (1+y)\big((1-p^{-s})^{-2} - 1\big)$, which proves (3.2). □

3.2 Factorization isolating the singularity

The Riemann zeta function admits the Euler product $\zeta(s) = \prod_p (1-p^{-s})^{-1}$ for $\mathrm{Re}(s) > 1$, so that $\zeta(s)^{2+2y} = \prod_p (1-p^{-s})^{-(2+2y)}$ on the same half-plane, provided a branch of the logarithm is fixed. We use the principal branch, which is analytic in s in a neighborhood of any point with $\mathrm{Re}(s) > 1$ and in y in a neighborhood of 0.

Define

$$H(s,y) := \prod_p (1-p^{-s})^{2+2y} \left(1 + (1+y)\left((1-p^{-s})^{-2} - 1\right)\right). \tag{3.3}$$

Theorem 3.2. For $\mathrm{Re}(s) > 1$ and y in a neighborhood of 0,

$$F(s,y) = \zeta(s)^{2+2y} H(s,y). \tag{3.4}$$

Proof. This follows by combining Lemma 3.1 with the Euler product for $\zeta(s)^{2+2y}$ and regrouping factors. □

The purpose of the factorization (3.4) is to isolate the singular behavior of $F(s,y)$ at $s = 1$ into the factor $\zeta(s)^{2+2y}$, leaving $H(s,y)$ as a well-behaved function in a larger region, as we now establish.

3.3 Analytic properties of $H(s,y)$

We show that the Euler product (3.3) defines a holomorphic function on a strip containing $\mathrm{Re}(s) = 1$, uniformly for y in a small disk around 0. The key is that each local factor $H_p(s,y)$ deviates from 1 by a quantity of order $p^{-2\,\mathrm{Re}(s)}$.

Lemma 3.3. Fix $\eta \in (0, 1/2)$ and $\delta \in (0, 1/2)$. Write $u := p^{-s}$ and denote the local factor by

$$H_p(s, y) := (1-u)^{2+2y} \left(1 + (1+y)\left((1-u)^{-2} - 1\right)\right).$$

Then for $|y| \le \eta$ and $\mathrm{Re}(s) \ge 1 - \delta$,

$$H_p(s, y) = 1 + O_{\eta,\delta}(u^2), \tag{3.5}$$

where the implied constant depends only on η and δ.

Proof. The condition $\mathrm{Re}(s) \ge 1 - \delta$ implies that, for every prime $p \ge 2$,

$$|u| = p^{-\mathrm{Re}(s)} \le p^{-(1-\delta)} \le 2^{-(1-\delta)} < 1,$$

since $\delta < 1/2$. In particular, $|u|$ is bounded away from 1, and the power series expansions

$$(1-u)^{-2} - 1 = 2u + 3u^2 + 4u^3 + \cdots = 2u + O_\delta(u^2),$$

$$(1-u)^{2+2y} = 1 - (2+2y)u + \binom{2+2y}{2}u^2 + \cdots = 1 - (2+2y)u + O_{\eta,\delta}(u^2),$$

converge absolutely, with remainders uniform for $|y| \le \eta$ and $\mathrm{Re}(s) \ge 1 - \delta$. Multiplying the expansions:

$$H_p(s, y) = \left(1 - (2+2y)u + O_{\eta,\delta}(u^2)\right)\left(1 + (1+y)(2u + O_\delta(u^2))\right).$$

The coefficient of u^1 on the right is $-(2+2y) + 2(1+y) = 0$. Hence $H_p(s, y) = 1 + O_{\eta,\delta}(u^2)$, proving (3.5). ☐

Proposition 3.4. Fix $\eta \in (0, 1/2)$ and $\delta \in (0, 1/2)$, and let

$$\mathcal{R}_{\delta,\eta} := \{(s, y) \in \mathbb{C}^2 : \mathrm{Re}(s) \ge 1 - \delta,\ |y| \le \eta\}. \tag{3.6}$$

Then the partial products of the Euler product (3.3) converge absolutely and uniformly on $\mathcal{R}_{\delta,\eta}$ to a function $H(s, y)$. For every y with $|y| \le \eta$, the function $H(s, y)$ is holomorphic in s on the open strip $\mathrm{Re}(s) > 1 - \delta$. Moreover, $H(s, y)$ is jointly holomorphic in (s, y) on $\mathrm{int}(\mathcal{R}_{\delta,\eta})$, and H is uniformly bounded on $\mathcal{R}_{\delta,\eta}$.

Proof. By Lemma 3.3, for every $(s, y) \in \mathcal{R}_{\delta,\eta}$ and every prime p,

$$|H_p(s, y) - 1| \le C_{\eta,\delta}\, p^{-2(1-\delta)}, \tag{3.7}$$

where the constant $C_{\eta,\delta}$ depends only on η and δ. Since $\delta < 1/2$, we have $2(1-\delta) > 1$, and the series $\sum_n n^{-2(1-\delta)}$ converges; in particular, $\sum_p p^{-2(1-\delta)}$ converges. By the Weierstrass M-test, the series $\sum_p (H_p(s, y) - 1)$ converges absolutely and uniformly on $\mathcal{R}_{\delta,\eta}$. By the standard theorem on infinite products—which states that an infinite product $\prod_n (1 + u_n)$ converges absolutely if and only if $\sum_n u_n$ does, and that the convergence of the partial products is uniform on a set whenever the corresponding series converges uniformly there—the partial products $\prod_{p \le P} H_p(s, y)$ converge absolutely and uniformly on $\mathcal{R}_{\delta,\eta}$ to a limit, which we denote $H(s, y)$.

To deduce holomorphy in s, fix $y_0 \in \mathbb{C}$ with $|y_0| \le \eta$ and let $s_0 \in \mathbb{C}$ be any point with $\mathrm{Re}(s_0) > 1 - \delta$. Choose $r > 0$ small enough that $r < \mathrm{Re}(s_0) - (1 - \delta)$, and define

$$K := \overline{D}(s_0, r) = \{s \in \mathbb{C} : |s - s_0| \le r\}. \tag{3.8}$$

The set K is closed and bounded in $\mathbb{C}$, so by the Heine–Borel theorem it is compact. By the choice of r, every $s \in K$ satisfies

$$\operatorname{Re}(s) \geq \operatorname{Re}(s_0) - r > 1 - \delta,$$

and hence $K \times \{y_0\} \subset \mathcal{R}_{\delta,\eta}$. The uniform convergence of the partial products on $\mathcal{R}_{\delta,\eta}$ established above therefore restricts to uniform convergence of $\prod_{p \leq P} H_p(s, y_0)$ on K as $P \to \infty$.

Each finite partial product $\prod_{p \leq P} H_p(s, y_0)$ is holomorphic in s on the open disk $D(s_0, r) = \operatorname{int}(K)$, since each factor $H_p(\cdot, y_0)$ is. By the theorem on uniform limits of holomorphic functions (Weierstrass), the limit $H(\cdot, y_0)$ is holomorphic on $D(s_0, r)$, and in particular at s_0. Since s_0 with $\operatorname{Re}(s_0) > 1 - \delta$ was arbitrary, $H(\cdot, y_0)$ is holomorphic on the open strip $\operatorname{Re}(s) > 1 - \delta$. Since y_0 with $|y_0| \leq \eta$ was also arbitrary, the holomorphy claim in s is proved.

For joint holomorphy, each finite partial product $\prod_{p \leq P} H_p(s, y)$ is holomorphic in (s, y) on $\operatorname{int}(\mathcal{R}_{\delta,\eta})$, since each factor $H_p(s, y)$ is. The uniform convergence of the partial products on $\mathcal{R}_{\delta,\eta}$ established above restricts to uniform convergence on every compact subset of $\operatorname{int}(\mathcal{R}_{\delta,\eta})$, since $\operatorname{int}(\mathcal{R}_{\delta,\eta}) \subset \mathcal{R}_{\delta,\eta}$. By the theorem on uniform limits of holomorphic functions in two complex variables, the limit $H(s, y)$ is jointly holomorphic on $\operatorname{int}(\mathcal{R}_{\delta,\eta})$.

Uniform boundedness on $\mathcal{R}_{\delta,\eta}$ follows from (3.7):

$$|H(s, y)| \leq \prod_p \big(1 + |H_p(s, y) - 1|\big) \leq \exp\left(\sum_p |H_p(s, y) - 1|\right) \leq \exp\left(C_{\eta,\delta} \sum_p p^{-2(1-\delta)}\right),$$

the right-hand side being finite and independent of $(s, y) \in \mathcal{R}_{\delta,\eta}$. □

Proposition 3.5. The function $H(s, y)$ satisfies

$$H(s, 0) \equiv 1 \quad \text{on } \operatorname{Re}(s) > 1/2. \tag{3.9}$$

In particular, $H(1, 0) = 1$.

Proof. On $\operatorname{Re}(s) > 1$, the factorization (3.4) specialized at $y = 0$ reads

$$F(s, 0) = \zeta(s)^2 \, H(s, 0).$$

On the other hand, the Dirichlet series defining F gives

$$F(s, 0) = \sum_{n \geq 1} \frac{d(n)}{n^s} = \zeta(s)^2 \quad \text{on } \operatorname{Re}(s) > 1.$$

Comparing the two expressions yields $\zeta(s)^2 H(s, 0) = \zeta(s)^2$ on $\operatorname{Re}(s) > 1$. Since $\zeta(s) \neq 0$ on $\operatorname{Re}(s) > 1$ (which follows directly from the Euler product representation of ζ), we may divide both sides by $\zeta(s)^2$ to obtain

$$H(s, 0) \equiv 1 \quad \text{on } \operatorname{Re}(s) > 1.$$

By Proposition 3.4, applied with any $\delta \in (0, 1/2)$, the function $H(s, 0)$ is holomorphic on $\operatorname{Re}(s) > 1 - \delta$; letting $\delta \to 1/2$ shows that $H(s, 0)$ is holomorphic on the connected open set $\operatorname{Re}(s) > 1/2$. The constant function 1 is also holomorphic there, and the two functions agree on the open subset $\operatorname{Re}(s) > 1$. By the identity theorem, they agree on all of $\operatorname{Re}(s) > 1/2$. In particular, $H(1, 0) = 1$. □

4 Perron formula and contour deformation

To extract the asymptotic behavior of $T_j(x)$ from the Dirichlet series $F(s,y)$, we work with the generating partial sum

$$A(x,y) := \sum_{n\le x} d(n)(1+y)^{\omega(n)}, \tag{4.1}$$

which satisfies the generating relation

$$A(x,y) = \sum_{j\ge 0} T_j(x) y^j. \tag{4.2}$$

Throughout this section, we initially fix one parameter:

- $\eta \in (0, 1/2)$, controlling the size of the disk $|y| \le \eta$.

After the contour depth δ' is fixed, we may shrink η if necessary for the contour estimates.

The fixed strip depth δ' and the truncation-dependent quantity λ_T will be chosen together in Section 4.3. This keeps the Euler factor estimates, the zero-free region, and the contour growth bound tied to the same strip parameter.

We also abbreviate $\alpha := 2 + 2y$. Thus the factorization (3.4) becomes $F(s,y) = \zeta(s)^\alpha H(s,y)$. For $|y| \le \eta$ with $\eta < 1/2$, the exponent α ranges in a disk of radius $2\eta < 1$ around 2, so in particular α is bounded away from non-positive integers.

4.1 The truncated Perron formula

We quote the following standard form of the truncated Perron formula (see, e.g., [, Theorem II.2.2]).

Theorem 4.1 (Truncated Perron formula)**.** Let $f(s) = \sum_{n\ge 1} a_n n^{-s}$ be a Dirichlet series with abscissa of absolute convergence σ_a. For real numbers $c > \sigma_a$, $x \ge 2$, and $T \ge 1$ with $x \notin \mathbb{Z}$,

$$\sum_{n\le x} a_n = \frac{1}{2\pi i}\int_{c-iT}^{c+iT} f(s)\frac{x^s}{s}\,ds + R(x,T), \tag{4.3}$$

where the truncation error satisfies

$$R(x,T) \ll \frac{x^c}{T}\sum_{n\ge 1}\frac{|a_n|}{n^c\left(1+T\left|\log\frac{x}{n}\right|\right)} + \frac{x^c}{T\log x}. \tag{4.4}$$

If $x = N$ is an integer, we do not apply the Perron formula directly at N. Instead, we apply it at

$$X := N + \frac{1}{2}.$$

Then $X \notin \mathbb{Z}$ and

$$A(N,y) = A(X,y),$$

because the integers satisfying $n \le N$ are exactly the integers satisfying $n \le N + 1/2$. Moreover,

$$X = N + O(1), \qquad \log X = \log N + O(1/N),$$

so every asymptotic estimate obtained for X transfers back to N. Thus it is enough in the Perron argument to work with non-integral x; the integer case follows from this replacement.

We apply Theorem 4.1 to the Dirichlet series $F(s,y) = \sum_{n\ge 1} d(n)(1+y)^{\omega(n)} n^{-s}$. For $|y| \le \eta$ with η sufficiently small, the series converges absolutely for $\mathrm{Re}(s) > 1$, so we may take any $c > 1$. Choose, for instance, $c = 1 + 1/\log x$, which is the standard choice to balance the main term size and the truncation error.

Proposition 4.2. For $x \geq 2$ with $x \notin \mathbb{Z}$, $|y| \leq \eta$, and $T \geq 1$,

$$A(x,y) = \frac{1}{2\pi i} \int_{c-iT}^{c+iT} F(s,y) \frac{x^s}{s} \, ds + E_{\text{Perron}}(x,T,y), \tag{4.5}$$

where $c = 1 + 1/\log x$ and

$$E_{\text{Perron}}(x,T,y) \ll \frac{x^c}{T} F(c,\eta) + \frac{x^c}{T \log x}. \tag{4.6}$$

Proof. Apply Theorem 4.1 with $a_n = d(n)(1+y)^{\omega(n)}$ and $c = 1 + 1/\log x$. Denote the resulting truncation error by $E_{\text{Perron}}(x,T,y)$. The error term in (4.4) gives

$$E_{\text{Perron}}(x,T,y) \ll \frac{x^c}{T} \sum_{n \geq 1} \frac{|d(n)(1+y)^{\omega(n)}|}{n^c \left(1 + T \left|\log \frac{x}{n}\right|\right)} + \frac{x^c}{T \log x}.$$

Since $|1+y| \leq 1 + \eta$, we have

$$|d(n)(1+y)^{\omega(n)}| \leq d(n)(1+\eta)^{\omega(n)}.$$

Using only the trivial inequality

$$1 + T \left|\log \frac{x}{n}\right| \geq 1,$$

we get

$$\begin{aligned} E_{\text{Perron}}(x,T,y) &\ll \frac{x^c}{T} \sum_{n \geq 1} \frac{d(n)(1+\eta)^{\omega(n)}}{n^c} + \frac{x^c}{T \log x} \\ &= \frac{x^c}{T} F(c,\eta) + \frac{x^c}{T \log x}, \end{aligned}$$

which is (4.6). □

4.2 Analytic preliminaries

To control the contour integrals arising from the deformation in Section 4.3, we will need:

(a) a region in which $\zeta(s)$ does not vanish, so that the contour can avoid its zeros;

(b) a continuous branch of $\zeta(s)^{2+2y}$ on a neighborhood of the contour;

(c) a uniform bound on $|\zeta(s)|$ along vertical lines in the critical strip;

(d) a corresponding bound on $\arg \zeta(s)$ on the chosen branch.

We collect these analytic ingredients here, before constructing the contour itself in Section 4.3.

Zero-free region

For the contour deformation to be valid, we need a region in which $\zeta(s) \neq 0$. We use the following classical result of de la Vallée Poussin, quoted without proof; see, e.g., [, Ch. 3] or [, §6.6].

Theorem 4.3 (de la Vallée Poussin zero-free region)**.** There exists an absolute constant $c_0 \in (0, 1/2)$ such that $\zeta(s) \neq 0$ on the open region

$$\mathcal{Z} := \left\{ s = \sigma + it \in \mathbb{C} : \sigma > 1 - \frac{c_0}{\log(|t|+2)} \right\} \setminus \{1\}. \tag{4.7}$$

The boundary of $\mathcal{Z}$ is curved, and integrating along it would complicate the estimation of the resulting contour pieces. We will instead use a fixed vertical line lying strictly inside $\mathcal{Z}$ for all $|t| \leq T$; the precise depth of this line is fixed in Section 4.3.

Continuous branch of $\zeta(s)^\alpha$

To make sense of $\zeta(s)^\alpha = \zeta(s)^{2+2y}$ on a neighborhood of the contour, we specify a branch on the region

$$\mathcal{D} := \mathcal{Z} \setminus \mathcal{S}, \tag{4.8}$$

where $\mathcal{S} := \{s \in \mathbb{C} : \Im(s) = 0,\ \mathrm{Re}(s) \le 1\}$ is the slit along the real axis to the left of $s = 1$. The slit $\mathcal{S}$ disconnects the segment $\mathcal{Z} \cap \mathcal{S}$ from the rest of $\mathcal{Z}$ from above and below, rendering $\mathcal{D}$ simply connected.[1]

On $\mathcal{D}$, both $\zeta(s)$ and $\zeta'(s)$ are holomorphic, and $\zeta(s) \neq 0$. Fix any real number $c_* > 1$ (for example $c_* = 2$); since $\zeta(c_*) > 0$, the principal logarithm $\log \zeta(c_*) \in \mathbb{R}$ is well-defined. Define

$$L(s) := \log \zeta(c_*) + \int_{c_*}^{s} \frac{\zeta'(\omega)}{\zeta(\omega)}\, d\omega \qquad (s \in \mathcal{D}), \tag{4.9}$$

where the integral is taken along any path in $\mathcal{D}$ from c_* to s. Since $\mathcal{D}$ is simply connected and ζ'/ζ is holomorphic on $\mathcal{D}$, the integral is path-independent, and L is a well-defined holomorphic function on $\mathcal{D}$.

We claim $e^{L(s)} = \zeta(s)$ on $\mathcal{D}$. Indeed, $L'(s) = \zeta'(s)/\zeta(s)$ on $\mathcal{D}$, so

$$\frac{d}{ds}\big(e^{-L(s)}\zeta(s)\big) = e^{-L(s)}\big(\zeta'(s) - L'(s)\zeta(s)\big) = 0.$$

Hence $e^{-L(s)}\zeta(s)$ is constant on $\mathcal{D}$, and evaluating at $s = c_*$ gives $e^{-L(c_*)}\zeta(c_*) = e^{-\log\zeta(c_*)}\zeta(c_*) = 1$. Therefore $e^{L(s)} = \zeta(s)$ throughout $\mathcal{D}$, so L serves as a continuous branch of $\log \zeta$ on $\mathcal{D}$.

We define

$$\zeta(s)^\alpha := e^{\alpha L(s)} \qquad (s \in \mathcal{D},\ \alpha \in \mathbb{C}). \tag{4.10}$$

This is single-valued and holomorphic in s on $\mathcal{D}$, and entire in α. At $\alpha = 2$, the definition reduces to $\zeta(s)^2$ in the usual sense.

Vertical growth of $\zeta(s)$

We use the following uniform bound on $|\zeta(s)|$, obtained by combining the convexity bound with the trivial estimate $\zeta(\sigma + it) \ll \log|t|$ for $\sigma \ge 1$ (see, e.g., [, Theorem II.3.8] for the convexity bound).

Theorem 4.4 (Uniform bound on $\zeta(s)$ in a vertical strip)**.** Fix $\delta \in (0, 1/2)$. For every $\varepsilon > 0$,

$$|\zeta(\sigma + it)| \ll_{\delta,\varepsilon} |t|^{\delta/2+\varepsilon} \quad \text{for } 1-\delta \le \sigma \le 1+\delta \text{ and } |t| \ge 2, \tag{4.11}$$

where the implied constant is independent of t.

Sketch. For $1 - \delta \le \sigma \le 1$, the convexity bound gives $|\zeta(\sigma+it)| \ll_\varepsilon |t|^{(1-\sigma)/2+\varepsilon}$. Taking the supremum of $(1-\sigma)/2$ over $\sigma \in [1-\delta, 1]$ yields $\delta/2$, so $|\zeta(\sigma+it)| \ll_\varepsilon |t|^{\delta/2+\varepsilon}$ on this subrange.

For $1 \le \sigma \le 1+\delta$, the standard estimate $|\zeta(\sigma+it)| \ll \log|t|$ holds, and $\log|t| \ll_\varepsilon |t|^\varepsilon \le |t|^{\delta/2+\varepsilon}$. Combining the two ranges gives (4.11). $\square$

[1] Indeed, $\mathcal{Z}$ is a slit-once-punctured domain that is not simply connected, since a small loop around $s = 1$ inside $\mathcal{Z}$ is not nullhomotopic. Removing the entire ray $(-\infty, 1]$ (equivalently, the connected component $\mathcal{Z} \cap \mathcal{S}$ of $\mathcal{S}$ inside $\mathcal{Z}$, which contains $s = 1$) cuts every such loop, making the resulting domain simply connected.

Bound on $\arg\zeta$

We also use the following standard estimate on the branch determined by (4.9); see, e.g., [, §9.4] or [, Theorem II.3.10].

Theorem 4.5 (Bound on $\arg\zeta$)**.** There exists an absolute constant $C_{\arg} > 0$ such that, for every $s = \sigma + it \in \mathcal{D}$ with $|t| \geq 2$,

$$|\arg\zeta(s)| \leq C_{\arg}\log|t|, \tag{4.12}$$

where the branch of $\arg\zeta$ is the one determined by the logarithm L on $\mathcal{D}$.

4.3 Contour decomposition

We now introduce the parameters governing the contour deformation and construct the deformed contour.

Depth of the deformation

Fix once and for all an absolute constant

$$\delta' \in \left(\frac{c_0}{2\log 3}, \frac{c_0}{2}\right), \tag{4.13}$$

where c_0 is the constant of Theorem 4.3; the interval is non-empty since $\log 3 > 1$. This range of δ' is chosen so that the deformed contour remains in the fixed strip used in Section 4.4.

For each truncation height $T \geq 1$, define

$$\lambda_T := \frac{c_0}{2\log(T+2)}. \tag{4.14}$$

By construction, $\lambda_T \to 0$ as $T \to \infty$, and for every $T \geq 1$,

$$\lambda_T \leq \frac{c_0}{2\log 3} < \delta', \tag{4.15}$$

so the depth $\sigma_0(T) := 1 - \lambda_T$ of the deformed contour satisfies $\sigma_0(T) > 1 - \delta'$. Consequently, the contour $\mathcal{C}(T,\rho)$ defined below lies in $\mathcal{R}_{\delta',\eta}$ for every $T \geq 1$ and every $|y| \leq \eta$. In particular, Proposition 3.4 applies on the contour with $\delta = \delta'$.

Finally, the choice $c = 1 + 1/\log x$ in Proposition 4.2 satisfies $c \leq 1 + \delta'$ for all x sufficiently large, ensuring that the abscissa c lies inside the strip $1 - \delta' \leq \sigma \leq 1 + \delta'$ used in the contour estimates. We assume this hereafter.

The deformed contour

Fix a truncation height $T \geq 1$, and set

$$\sigma_0 := \sigma_0(T) := 1 - \lambda_T. \tag{4.16}$$

Choose a parameter $\rho > 0$ with $\rho < \lambda_T$; its precise choice as a function of x will be made in Section 5. The notation $i0^{\pm}$ below denotes boundary values: the two Hankel arms are first taken at heights $\pm\varepsilon$ with $\varepsilon > 0$, and the limit $\varepsilon \to 0^+$ is taken after the corresponding integrals are formed. Thus the approximating contours lie in the slit domain $\mathcal{D}$ before the boundary value limit is taken.

We deform the original Perron segment $[c - iT, c + iT]$ leftward, around the singularity at $s = 1$, into a contour $\mathcal{C} = \mathcal{C}(T,\rho)$ traversed from $c - iT$ to $c + iT$. It consists of the following five pieces:

(1) $\mathcal{C}_{\mathrm{bot}}$: the horizontal segment from $c - iT$ to $\sigma_0 - iT$.

(2) $\mathcal{C}_{\mathrm{left}}^{-}$: the vertical segment from $\sigma_0 - iT$ to $\sigma_0 + i0^{-}$.

(3) $\mathcal{H}_{\mathrm{loc}}$: a local Hankel contour around $s = 1$, consisting of

   (3a) the lower arm from $\sigma_0 + i0^{-}$ to $1 - \rho + i0^{-}$;

   (3b) the circle $|s - 1| = \rho$, traversed from $1 - \rho + i0^{-}$ to $1 - \rho + i0^{+}$ counterclockwise through the right side of $s = 1$;

   (3c) the upper arm from $1 - \rho + i0^{+}$ to $\sigma_0 + i0^{+}$.

(4) $\mathcal{C}_{\mathrm{left}}^{+}$: the vertical segment from $\sigma_0 + i0^{+}$ to $\sigma_0 + iT$.

(5) $\mathcal{C}_{\mathrm{top}}$: the horizontal segment from $\sigma_0 + iT$ to $c + iT$.

The five pieces are joined consecutively, so $\mathcal{C}$ has the same initial and terminal points as the original Perron contour. The Hankel arms are understood as boundary values along the slit $\mathcal{S}$; in the local representation near $s = 1$, their discontinuity is carried by the factor $(s - 1)^{-2-2y}$ and produces the local contribution I_H.

Verification that the contour avoids ζ-zeros

We claim that $\mathcal{C} \setminus \{1\} \subset \mathcal{Z}$, where $\mathcal{Z}$ is the zero-free region of Theorem 4.3. For the horizontal pieces $\mathcal{C}_{\mathrm{bot}}$ and $\mathcal{C}_{\mathrm{top}}$,

$$\mathrm{Re}(s) \geq \sigma_0(T) = 1 - \frac{c_0}{2\log(T+2)} > 1 - \frac{c_0}{\log(|\Im s| + 2)}. \tag{4.17}$$

For the vertical pieces $\mathcal{C}_{\mathrm{left}}^{\pm}$,

$$\mathrm{Re}(s) = \sigma_0(T) = 1 - \frac{c_0}{2\log(T+2)} > 1 - \frac{c_0}{\log(|\Im s| + 2)}. \tag{4.18}$$

For the local Hankel contour $\mathcal{H}_{\mathrm{loc}}$, the two horizontal arms (3a) and (3c) satisfy $\mathrm{Re}(s) \geq \sigma_0(T)$ and $|\Im s| \leq \rho$, hence are covered by the same estimate as the vertical pieces. For the small circle (3b),

$$\mathrm{Re}(s) \geq 1 - \rho > 1 - \lambda_T = 1 - \frac{c_0}{2\log(T+2)} > 1 - \frac{c_0}{\log(|\Im s| + 2)}. \tag{4.19}$$

Therefore $\mathcal{C} \setminus \{1\} \subset \mathcal{Z}$. Combined with the boundary-value convention for the Hankel arms, the approximating contours with heights $\pm\varepsilon$ are contained in $\mathcal{D}$ away from $s = 1$. The integrand $F(s, y)x^s/s$ is therefore well-defined and holomorphic on a neighborhood of each approximating contour, and the contour $\mathcal{C}$ is understood as the corresponding boundary-value limit.

Cauchy's theorem

The Perron contour and $\mathcal{C}$ share the same endpoints $c \pm iT$. Together, the Perron contour traversed from $c - iT$ to $c + iT$ and $\mathcal{C}$ traversed in the reverse direction form a closed loop Γ, oriented counterclockwise. The keyhole formed by $\mathcal{H}_{\mathrm{loc}}$ and the slit $\mathcal{S}$ excludes the point $s = 1$ from the interior of Γ, so the interior of Γ is contained in the rectangle

$$\mathcal{R}_\Gamma := \{s = \sigma + it : \sigma_0(T) \leq \sigma \leq c,\ |t| \leq T\} \setminus \{1\}, \tag{4.20}$$

with the slit further excluded. For every $s \in \mathcal{R}_\Gamma$,

$$\operatorname{Re}(s) \geq \sigma_0(T) = 1 - \frac{c_0}{2\log(T+2)} > 1 - \frac{c_0}{\log(|\Im s|+2)}, \tag{4.21}$$

so $\mathcal{R}_\Gamma \subset \mathcal{Z}$. Combined with the on-contour verifications (4.17)–(4.19), this shows that Γ together with its interior is contained in $\mathcal{Z}$. In particular, $F(s,y)x^s/s$ is holomorphic on a neighborhood of Γ and its interior, with the slit $\mathcal{S}$ removed. By Cauchy's theorem,

$$\oint_\Gamma F(s,y)\frac{x^s}{s}\,ds = 0, \tag{4.22}$$

which yields

$$\int_{c-iT}^{c+iT} F(s,y)\frac{x^s}{s}\,ds = \int_{\mathcal{C}} F(s,y)\frac{x^s}{s}\,ds. \tag{4.23}$$

Decomposition of $A(x,y)$

For non-integral x, combining (4.23) with the truncated Perron formula (Proposition 4.2), we obtain

$$A(x,y) = I_H + I_{\mathrm{top}} + I_{\mathrm{bot}} + I_{\mathrm{left}} + E_{\mathrm{Perron}}(x,T,y), \tag{4.24}$$

where, for $* \in \{\mathrm{top},\ \mathrm{bot},\ \mathrm{left}\}$,

$$I_* := \frac{1}{2\pi i}\int_{\mathcal{C}_*} F(s,y)\frac{x^s}{s}\,ds, \tag{4.25}$$

with $\mathcal{C}_{\mathrm{left}} := \mathcal{C}_{\mathrm{left}}^- \cup \mathcal{C}_{\mathrm{left}}^+$, and

$$I_H := \frac{1}{2\pi i}\int_{\mathcal{H}_{\mathrm{loc}}} F(s,y)\frac{x^s}{s}\,ds. \tag{4.26}$$

The local Hankel contribution I_H contains the main asymptotic behavior and will be evaluated in Section 5. The remaining contributions $I_{\mathrm{top}}, I_{\mathrm{bot}}, I_{\mathrm{left}}$ are estimated in Section 4.4 and shown to be of lower order than the main term, after a suitable choice of $T = T(x)$.

4.4 Estimates of the contour pieces

In this section, we estimate the three nonlocal contour contributions $I_{\mathrm{top}}, I_{\mathrm{bot}}, I_{\mathrm{left}}$ appearing in the decomposition (4.24), and combine them with the Perron truncation error E_{Perron} to obtain a single error bound. The local Hankel contribution I_H is deferred to Section 5.

We first collect the two bounds for $F(s,y)$ used below.

Lemma 4.6 (Bounds for $F(s,y)$ on the deformed contour)**.** Let $0 < \eta < 1/2$, and let δ' be fixed as in (4.13). Let $\varepsilon > 0$ and put $\alpha = 2 + 2y$. Define

$$B := B(\eta,\delta',\varepsilon) := (2+2\eta)\left(\tfrac{\delta'}{2} + \varepsilon\right) + 2\eta C_{\mathrm{arg}}. \tag{4.27}$$

Then the following estimates hold uniformly for $|y| \leq \eta$.

1. If $s = \sigma + it \in \mathcal{D}$, $|t| \geq 2$, and $1 - \delta' \leq \sigma \leq 1 + \delta'$, then

$$|F(s,y)| \ll_{\eta,\delta',\varepsilon} |t|^B. \tag{4.28}$$

2. If $T \geq 1$, $s = 1 - \lambda_T + it$, and $|t| \leq 2$, then

$$|F(s,y)| \ll_{\eta,\delta'} \lambda_T^{-2-2\eta} \ll_{\eta,\delta'} (\log(T+2))^{2+2\eta}. \tag{4.29}$$

Proof. For the first estimate, let $s = \sigma + it \in \mathcal{D}$ with $|t| \geq 2$ and $1 - \delta' \leq \sigma \leq 1 + \delta'$. On $\mathcal{D}$, the chosen branch gives

$$F(s,y) = \zeta(s)^\alpha H(s,y), \qquad \zeta(s)^\alpha = \exp(\alpha L(s)).$$

Writing $L(s) = \log|\zeta(s)| + i \arg \zeta(s)$ on this branch,

$$|\zeta(s)^\alpha| = |\zeta(s)|^{\mathrm{Re}(\alpha)} \exp\bigl(-\Im(\alpha) \arg \zeta(s)\bigr).$$

Since $|y| \leq \eta$, we have $\mathrm{Re}(\alpha) \leq 2 + 2\eta$ and $|\Im(\alpha)| \leq 2\eta$. Hence

$$|F(s,y)| \leq |H(s,y)|(1 + |\zeta(s)|)^{2+2\eta} \exp\bigl(2\eta|\arg \zeta(s)|\bigr).$$

Here the factor $1 + |\zeta(s)|$ avoids any lower-bound assumption on $|\zeta(s)|$. Proposition 3.4, applied with $\delta = \delta'$, gives $|H(s,y)| \ll_{\eta,\delta'} 1$. Moreover, Theorem 4.4 gives

$$|\zeta(s)| \ll_{\delta',\varepsilon} |t|^{\delta'/2+\varepsilon},$$

and Theorem 4.5 gives

$$|\arg \zeta(s)| \leq C_{\arg} \log |t|.$$

Since $|t| \geq 2$, these estimates imply

$$|F(s,y)| \ll_{\eta,\delta',\varepsilon} |t|^{(2+2\eta)(\delta'/2+\varepsilon)+2\eta C_{\arg}} = |t|^B,$$

which proves (4.28).

We now prove the low-height estimate. Let

$$q(s) := (s-1)\zeta(s), \qquad q(1) := 1.$$

Then q is holomorphic at $s = 1$. Put

$$\lambda_0 := \frac{c_0}{2\log 3}, \qquad K := \{s = \sigma + it : 1 - \lambda_0 \leq \sigma \leq 1,\ |t| \leq 2\}.$$

For $|t| \leq 2$ we have $\log(|t| + 2) \leq \log 4 < 2\log 3$, and hence

$$\lambda_0 = \frac{c_0}{2\log 3} < \frac{c_0}{\log(|t|+2)}.$$

Thus $K \setminus \{1\}$ lies in the zero-free region of Theorem 4.3. After setting $q(1) = 1$, the function q is holomorphic and non-vanishing on K. Since K is compact, there is an open neighborhood U of K on which q is non-vanishing. Choose a holomorphic branch

$$M(s) = \log q(s)$$

on U, compatible with $M(s) = L(s) + \mathrm{Log}(s-1)$ on $U \cap \mathcal{D}$. Let

$$M_K := \sup_{s \in K} |M(s)| < \infty.$$

For $|y| \le \eta$,

$$|q(s)^\alpha| = |\exp(\alpha M(s))| \le \exp(|\alpha|\,|M(s)|) \le \exp((2+2\eta)M_K) \qquad (s \in K).$$

Define

$$G(s,y) := q(s)^\alpha H(s,y). \tag{4.30}$$

Then, on the chosen branch near the slit,

$$F(s,y) = G(s,y)(s-1)^{-\alpha}. \tag{4.31}$$

Since $K \subset \{s : \mathrm{Re}(s) \ge 1-\delta'\}$, combining the preceding bound for $q(s)^\alpha$ with Proposition 3.4, again with $\delta = \delta'$, gives

$$|G(s,y)| = |q(s)^\alpha H(s,y)| \ll_{\eta,\delta'} 1 \qquad (s \in K,\ \ |y| \le \eta).$$

On the left vertical line $s = 1 - \lambda_T + it$ with $|t| \le 2$, we use (4.31). For either boundary value of the logarithm along the slit, $|\operatorname{Arg}(s-1)| \le \pi$, so

$$|(s-1)^{-\alpha}| = |s-1|^{-\mathrm{Re}(\alpha)} \exp\big(\Im(\alpha)\operatorname{Arg}(s-1)\big) \ll_\eta |s-1|^{-\mathrm{Re}(\alpha)}.$$

Since $|s-1| \ge \lambda_T$, if $|s-1| \le 1$ then

$$|s-1|^{-\mathrm{Re}(\alpha)} \le |s-1|^{-2-2\eta} \le \lambda_T^{-2-2\eta}.$$

If $|s-1| \ge 1$, then $\mathrm{Re}(\alpha) \ge 2 - 2\eta > 0$, so $|s-1|^{-\mathrm{Re}(\alpha)} \le 1 \le \lambda_T^{-2-2\eta}$. Hence

$$|F(1-\lambda_T + it, y)| \ll_{\eta,\delta'} \lambda_T^{-2-2\eta}.$$

The definition (4.14) gives

$$\lambda_T^{-2-2\eta} = \left(\frac{2\log(T+2)}{c_0}\right)^{2+2\eta} \ll_{\eta,\delta'} (\log(T+2))^{2+2\eta},$$

which proves (4.29). $\square$

Since

$$B(\eta,\delta',\varepsilon) = (2+2\eta)\left(\tfrac{\delta'}{2} + \varepsilon\right) + 2\eta C_{\arg} \longrightarrow \delta' \qquad (\eta,\varepsilon \to 0^+),$$

and $\delta' < c_0/2$, we may choose $\eta > 0$ and $\varepsilon > 0$ sufficiently small so that

$$B < c_0/2. \tag{4.32}$$

This choice is fixed throughout the contour estimates.

Nonlocal contour pieces

Lemma 4.7 (Bound for the horizontal contour pieces)**.** For all sufficiently large x and every $T \ge 2$,

$$|I_{\mathrm{top}}| + |I_{\mathrm{bot}}| \ll_{\eta,\varepsilon} x^c\, T^{B-1}, \tag{4.33}$$

uniformly in $|y| \le \eta$.

Proof. On $\mathcal{C}_{\mathrm{top}}$, parameterize $s = \sigma + iT$ with $\sigma_0(T) \le \sigma \le c$. Then

$$\begin{aligned}
|I_{\mathrm{top}}| &= \left| \frac{1}{2\pi i} \int_{\mathcal{C}_{\mathrm{top}}} F(s,y) \frac{x^s}{s}\, ds \right| \\
&\le \frac{1}{2\pi} \int_{\sigma_0(T)}^{c} \left| F(\sigma + iT, y) \frac{x^{\sigma+iT}}{\sigma + iT} \right| d\sigma \\
&\le \frac{c - \sigma_0(T)}{2\pi} \sup_{\sigma_0(T) \le \sigma \le c} \left| F(\sigma + iT, y) \frac{x^{\sigma+iT}}{\sigma + iT} \right|.
\end{aligned}$$

On this segment, $|x^{\sigma+iT}| = x^\sigma \le x^c$ and $|\sigma + iT| \ge T$. Also $\sigma_0(T) \ge 1 - \delta'$ by (4.15), while $c \le 1 + \delta'$ for all sufficiently large x. Hence Lemma 4.6 gives

$$\sup_{\sigma_0(T) \le \sigma \le c} |F(\sigma + iT, y)| \ll_{\eta,\varepsilon} T^B.$$

Since

$$c - \sigma_0(T) = \frac{1}{\log x} + \lambda_T = \frac{1}{\log x} + \frac{c_0}{2\log(T+2)} \ll 1,$$

we obtain

$$\begin{aligned}
|I_{\mathrm{top}}| &\ll_{\eta,\varepsilon} \frac{x^c}{T} \sup_{\sigma_0(T) \le \sigma \le c} |F(\sigma + iT, y)| \\
&\ll_{\eta,\varepsilon} x^c T^{B-1}.
\end{aligned}$$

The same computation on $\mathcal{C}_{\mathrm{bot}}$, with $s = \sigma - iT$, gives

$$|I_{\mathrm{bot}}| \ll_{\eta,\varepsilon} x^c T^{B-1}.$$

This proves (4.33). □

Lemma 4.8 (Bound for the left vertical contour piece)**.** For every $T \ge 2$,

$$|I_{\mathrm{left}}| \ll_{\eta,\varepsilon} x^{1-\lambda_T}\, T^B, \tag{4.34}$$

uniformly in $|y| \le \eta$.

Proof. The contour $\mathcal{C}_{\mathrm{left}} = \mathcal{C}_{\mathrm{left}}^- \cup \mathcal{C}_{\mathrm{left}}^+$ is parameterized by $s = \sigma_0(T) + it$, $-T \le t \le T$. Using $\sigma_0(T) = 1 - \lambda_T$, we first write

$$\begin{aligned}
|I_{\mathrm{left}}| &= \left| \frac{1}{2\pi i} \int_{\mathcal{C}_{\mathrm{left}}} F(s,y) \frac{x^s}{s}\, ds \right| \\
&\le \frac{1}{2\pi} \int_{-T}^{T} \left| F(\sigma_0(T) + it, y) \frac{x^{\sigma_0(T)+it}}{\sigma_0(T) + it} \right| dt \\
&= \frac{x^{1-\lambda_T}}{2\pi} \int_{-T}^{T} \left| \frac{F(\sigma_0(T) + it, y)}{\sigma_0(T) + it} \right| dt.
\end{aligned}$$

Since $\sigma_0(T) > 1 - \delta' > 1/2$, we have $|\sigma_0(T) + it| \gg 1$ for $|t| \le 2$ and $|\sigma_0(T) + it| \gg |t|$ for $2 \le |t| \le T$. Hence

$$|I_{\mathrm{left}}| \ll x^{1-\lambda_T} \left(\int_{|t| \le 2} |F(\sigma_0(T) + it, y)|\, dt + \int_{2 \le |t| \le T} \frac{|F(\sigma_0(T) + it, y)|}{|t|}\, dt \right).$$

On $|t| \le 2$, Lemma 4.6 gives

$$|F(\sigma_0(T)+it,y)| \ll_{\eta,\delta'} (\log(T+2))^{2+2\eta}.$$

On $2 \le |t| \le T$, the point $\sigma_0(T)+it$ lies in $\mathcal{D}$ as a boundary-value limit and satisfies $1-\delta' \le \sigma_0(T) \le 1+\delta'$, so Lemma 4.6 gives

$$|F(\sigma_0(T)+it,y)| \ll_{\eta,\varepsilon} |t|^B.$$

Therefore

$$\begin{aligned} |I_{\text{left}}| &\ll_{\eta,\delta',\varepsilon} x^{1-\lambda_T}\left((\log(T+2))^{2+2\eta} + \int_{2\le|t|\le T} |t|^{B-1}\,dt\right) \\ &\ll_{\eta,\delta',\varepsilon} x^{1-\lambda_T}\left((\log(T+2))^{2+2\eta} + 2\int_2^T t^{B-1}\,dt\right). \end{aligned}$$

Since $B>0$,

$$2\int_2^T t^{B-1}\,dt = \frac{2}{B}\big(T^B - 2^B\big) \ll_B T^B.$$

Also $(\log(T+2))^{2+2\eta} \ll_{\eta,B} T^B$ for $T \ge 2$. Combining these estimates gives

$$|I_{\text{left}}| \ll_{\eta,\delta',\varepsilon} x^{1-\lambda_T}T^B.$$

This proves (4.34). $\square$

Choice of T and the combined error

We choose

$$T = T(x) := \exp\big(\sqrt{\log x}\big), \tag{4.35}$$

the standard Selberg–Delange truncation height adapted to the de la Vallée Poussin zero-free region. With this choice, $\log T = \sqrt{\log x}$, and $\lambda_T = c_0/(2\sqrt{\log x}+O(1))$, so $\lambda_T < \delta'$ holds automatically. Moreover, $c = 1+1/\log x \le 1+\delta'$ for all x sufficiently large, so the abscissa c lies inside the strip used in the growth estimates. We have

$$x^{-\lambda_T} = \exp(-\lambda_T \log x) = \exp\left(-\tfrac{c_0}{2}\sqrt{\log x}\cdot(1+o(1))\right). \tag{4.36}$$

Lemma 4.9 (Perron truncation error)**.** For $x \ge 2$ with $x \notin \mathbb{Z}$, $|y| \le \eta$, $T \ge 1$, and $c = 1+1/\log x$, we have

$$E_{\text{Perron}}(x,T,y) \ll_{\eta,\delta'} \frac{x(\log x)^{2+2\eta}}{T}. \tag{4.37}$$

Proof. Starting from the raw Perron error bound (4.6),

$$E_{\text{Perron}}(x,T,y) \ll \frac{x^c}{T}F(c,\eta) + \frac{x^c}{T\log x}.$$

On the line $\mathrm{Re}(s) = c > 1$, the Dirichlet series is absolutely convergent and the factorization is controlled directly. For sufficiently large x, we have $c \le 1+\delta'$. Since

$$F(c,\eta) = \zeta(c)^{2+2\eta}H(c,\eta),$$

and $c = 1 + 1/\log x > 1$, we have

$$\zeta(c) \le \frac{c}{c-1} \ll \log x.$$

Also, by Proposition 3.4, applied with the fixed strip depth δ', we have

$$H(c, \eta) \ll_{\eta,\delta'} 1.$$

Therefore

$$F(c, \eta) \ll_{\eta,\delta'} (\log x)^{2+2\eta}.$$

For the remaining bounded range of x, the same estimate follows after enlarging the implied constant, because c then ranges over a compact subinterval of $(1, \infty)$ and $F(c, \eta)$ is bounded there by absolute convergence. Finally,

$$x^c = x^{1+1/\log x} = ex \asymp x.$$

Substituting these estimates into the raw Perron error bound gives

$$E_{\text{Perron}}(x, T, y) \ll_{\eta,\delta'} \frac{x(\log x)^{2+2\eta}}{T},$$

as claimed. □

Proposition 4.10. With $T = T(x) = \exp(\sqrt{\log x})$, there exists a constant $c_1 > 0$ (depending on η, ε but not on x) such that

$$E_{\text{Perron}}(x, T, y) + |I_{\text{top}}| + |I_{\text{bot}}| + |I_{\text{left}}| \ll_{\eta,\varepsilon} x \exp\bigl(-c_1 \sqrt{\log x}\bigr), \tag{4.38}$$

uniformly in $|y| \le \eta$, for all sufficiently large $x \notin \mathbb{Z}$.

Proof. We bound each term separately and take the worst.

Perron error. By Lemma 4.9 with $T = \exp(\sqrt{\log x})$,

$$E_{\text{Perron}} \ll_{\eta,\delta'} \frac{x(\log x)^{2+2\eta}}{T} = x(\log x)^{2+2\eta} \exp(-\sqrt{\log x}),$$

which is $\ll x \exp(-(1 - o(1))\sqrt{\log x})$.

Horizontal pieces. With $c = 1 + 1/\log x$ so that $x^c = e \cdot x$, by Lemma 4.7,

$$|I_{\text{top}}| + |I_{\text{bot}}| \ll_{\eta,\varepsilon} x \cdot T^{B-1} = x \cdot \exp\bigl((B-1)\sqrt{\log x}\bigr).$$

Since $B < 1$, this is $\ll x \exp(-(1 - B)\sqrt{\log x})$.

Vertical piece. By Lemma 4.8 and (4.36),

$$|I_{\text{left}}| \ll_{\eta,\varepsilon} x \cdot x^{-\lambda_T} \cdot T^B = x \cdot \exp\bigl(-\tfrac{c_0}{2}\sqrt{\log x} + B\sqrt{\log x}\bigr)(1 + o(1)).$$

By the choice (4.13) of δ' together with (4.32), we have $B < c_0/2$. Hence $|I_{\text{left}}| \ll x \exp(-(c_0/2 - B)\sqrt{\log x})$.

Set

$$c_1 := \frac{1}{2}\left(\frac{c_0}{2} - B\right) > 0.$$

Since $0 < c_0 < 1/2$ and $B > 0$, the constant $c_0/2 - B$ is smaller than both 1 and $1 - B$. Therefore, for all sufficiently large x, the preceding three estimates are bounded by the right-hand side of (4.38). □

The right-hand side of (4.38) is $o(x \log x)$, so it will be dominated by the local Hankel contribution I_H to be analyzed in Section 5, which is of order $x \log x$.

5 Local Hankel analysis

6 Extraction of $T_j(x)$ and proof of the main theorem

7 Toward $D(x)$: outlook

References